

Multi-person Physics-based Pose Estimation for Combat Sports

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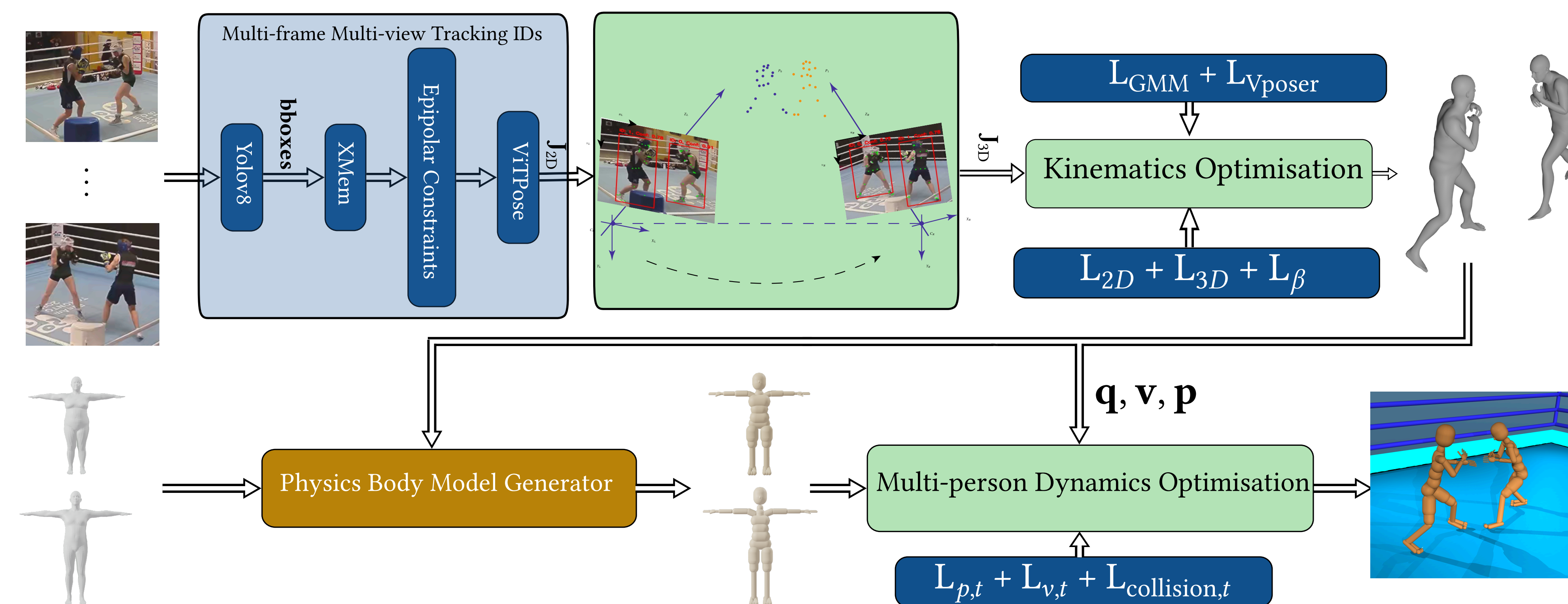
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Introduction

Motion capture in sports scenes like boxing presents challenges due to occlusion, crowding, and fast movements. We use a multi-view setup that requires visibility from only two cameras. Our pipeline produces realistic motions through multi-person physics optimisation while accounting for inter-character collisions.

Pipeline



The pipeline begins with bounding boxes (**bboxes**) and robust tracking IDs. 2D poses ($\mathbf{J}_{2D}$) from ViTPose [5] are triangulated to 3D ($\mathbf{J}_{3D}$). Kinematics optimisation fits SMPL (θ, β), then an iLQR physics stage [2] corrects artefacts using state $(\mathbf{q}, \mathbf{v}, \mathbf{p})$.

Multi-frame Multi-view Tracking IDs

- **Tracking by segmentation:** XMem memory for stable IDs [1]
- **Epipolar ID matching:** Centroid & epipolar constraints [4]
- **Top-down 2D poses:** ViTPose inside each box [5]

Weighted Triangulation and Filtering

Weighted triangulation solves

$$\mathbf{A} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0, \quad \mathbf{A} = \begin{bmatrix} \mu_1(P_{11} - u_1 P_{31}) \\ \mu_1(P_{21} - v_1 P_{31}) \\ \vdots \\ \mu_N(P_{1N} - u_N P_{31N}) \\ \mu_N(P_{2N} - v_N P_{32N}) \end{bmatrix}.$$

SVD yields (x, y, z) . Cubic-spline smoothing removes outliers; an EKF fills missing frames when triangulation fails.

Kinematics Optimization

Our kinematics optimization refines the SMPL model to minimize the difference between the provided 2D poses ($\mathbf{J}_{2D}$) from multiple views and 3D keypoints ($\mathbf{J}_{3D}$) obtained from triangulation. The process ensures temporal coherency and natural motion by incorporating smoothness and human motion priors. An LBFGS optimizer is used to solve the minimization problem, considering various loss terms, including 2D re-projection, 3D alignment, smoothness, and prior losses.

Kinematics Optimization Formulation: $\min_{\theta} w_1 L_{2D} + w_2 L_{3D} + w_3 L_{\text{reg}} + w_4 L_{\text{smooth}} + w_5 L_{\text{GMM}} + w_6 L_{\text{Vposer}}$

- **2D Re-projection Loss:** Aligns the 3D joints with 2D joints using re-projection, weighted by confidence.

$$L_{2D} = \sum_{j \in \mathcal{J}'} \sum_{i \in \mathcal{I}_{2D}} c_{j,i} \rho(\mathbf{J}_{\text{proj},ji} - \mathbf{J}_{2D,ji})$$
- **3D Alignment Loss:** Minimizes the difference between predicted and actual 3D joint positions considering confidence.

$$L_{3D} = \sum_{i \in \mathcal{J}_{3D}} c_i \|\mathbf{J}_i(\theta, \beta) - \mathbf{J}_{3D,i}\|^2$$
- **Smoothness Loss:** Ensures temporal consistency by minimizing differences in poses and vertices over time.

$$L_{\text{smooth}} = \sum \|\theta^t - \theta^{t-1}\|^2 + \|\mathcal{M}(\theta^t, \beta) - \mathcal{M}(\theta^{t-1}, \beta)\|^2$$
- **Prior Loss:** Guides the optimization towards natural poses using GMM and Vposer priors.

$$L_{\text{GMM}} = \frac{1}{N} \sum_{i=1}^N \text{GMM}(\theta_i, \beta), \quad L_{\text{Vposer}} = \frac{1}{N} \sum_{i=1}^N (z(\theta_i)^2)$$

Physics Body Model Generator

For K humanoids the state is $\mathbf{x}_t = (\mathbf{q}_t, \mathbf{v}_t) \in \mathbb{R}^{125K}$, with $\mathbf{p}_t \in \mathbb{R}^{21K}$ SMPL joints. We export to MuJoCo XML [3].

Multi-person Dynamics Optimization

The iLQR algorithm [2] optimizes a control trajectory $\mathbf{u}_{0:T}$ by minimizing the cost function. The reference positions $\mathbf{p}_t$ and velocities $\mathbf{v}_t$ are extracted from the kinematics optimization at each frame t , and the trajectory optimization is "warm-started" using control parameters $\mathbf{u}_{t-1}$ from the previous frame. The control is updated iteratively as $\Delta \mathbf{u}_t = \mathbf{K}_t \Delta \mathbf{x}_t + \alpha \mathbf{k}_t$, with $\mathbf{K}_t$ and $\mathbf{k}_t$ refined until convergence.

Dynamics Optimization Formulation: $\min_{\mathbf{u}_{0:T}} \sum_{t \in [0,T]} w_1 L_{\text{reg},t} + w_2 L_{\text{p},t} + w_3 L_{\text{v},t} + w_4 L_{\text{collision},t}$

- **Position Loss:** Minimizes the difference between predicted and actual 3D joint positions in each short horizon.

$$L_{\text{p},t} = \sum_{i \in \mathcal{J}_{3D}} \|\mathbf{p}_{i,t} - \mathbf{J}_{3D,i,t}\|^2$$
- **Velocity Loss:** Minimizes the difference between predicted and actual 3D joint velocities in each short horizon.

$$L_{\text{v},t} = \sum_{i \in \mathcal{J}_{3D}} c_i \|\mathbf{v}_{i,t} - \mathbf{v}_{3D,i,t}\|^2$$
- **Collision Loss:** The collision term penalizes the penetration between the geoms of different humanoids.

$$L_{\text{collision},t} = \sum_{i \in 1 \dots M} \sum_{j \in 1 \dots M} \max(0, \epsilon - \text{dist}(G_i, G_j))$$
- **Regularization Loss:** Prevent sudden changes in control input. $L_{\text{reg},t} = \|\mathbf{u}_t\|^2$

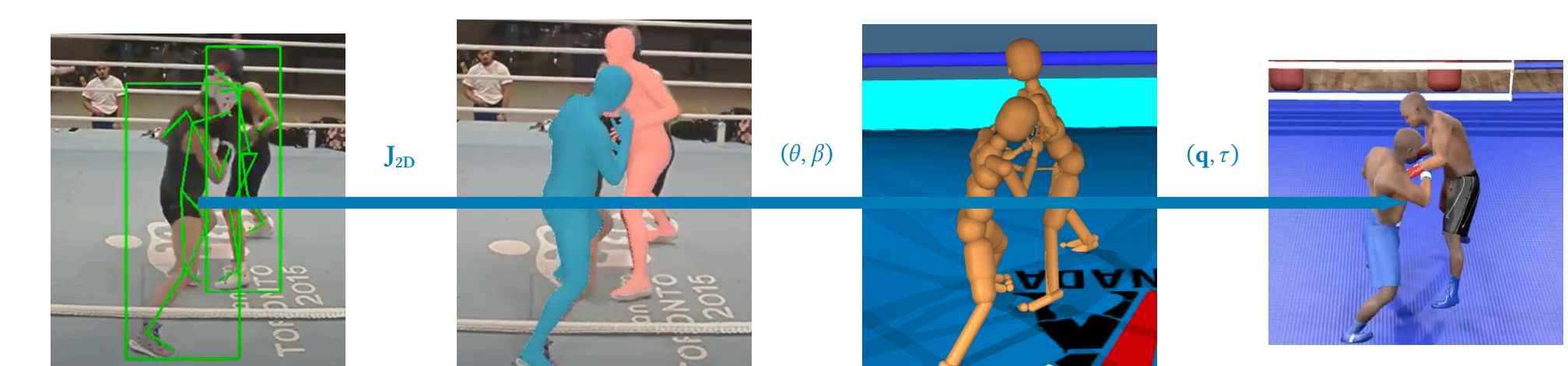
Discussion and Future Works

We captured a 10-hour boxing dataset to enable real-time inference and richer contact modeling.

Video



Visualisation



References